Triplet Adversarial Domain Adaptation for Pixel-Level Classification of VHR Remote Sensing Images

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***Abstract*— Pixel-level classification for very high resolution (VHR) images is a crucial but challenging task in remote sensing. However, since the diverse ways of satellite image acquisition and the distinct structures of various regions, the distributions of the same semantic classes among different data sets are dissimilar. Therefore, the classification model trained on one data set (source domain) may collapse, when it is directly applied to another one (target domain). To solve this problem, many adversarial-based domain adaptation methods have been proposed. However, these methods only consider the source and the target domains independently in the adversarial training, where only the target domain is explicitly contributed to narrow the gap between the distributions of both domains. Unlike previous methods, we propose a triplet adversarial domain adaptation (TriADA) method that jointly considers both domains to learn a domain-invariant classifier by a novel domain similarity discriminator. Specifically, the discriminator takes a triplet of segmentation maps as input, where two segmentation maps from the same domain are to be distinguished from the two maps from the different domains during the adversarial learning. Consequently, it explicitly considers both domains' information to narrow the distribution gap across domains. To enhance the discriminability of the classifier on the target domain, a class-aware self-training strategy, which depends on the output of the discriminator, is proposed to assign pseudo-labels with high adapted confidence on target data to retrain the classifier. Extensive experiments on several VHR pixel-level classification benchmarks demonstrate the effectiveness of our method as well as its superiority to the-state of the art.**

Manuscript received May 27, 2019; revised September 9, 2019; accepted December 2, 2019. Date of publication December 30, 2019; date of current version April 22, 2020. This work was supported in part by the Major Project for New Generation of Artificial Intelligence (AI) under Grant 2018AAA0100402 and in part by the National Natural Science Foundation of China under Grant 61573352, Grant 91646207, Grant 61620106003, and Grant 61773377, and in part by the Young Elite Scientists Sponsorship Program by China Association for Science and Technology (CAST) under Grant 2018QNRC001. *(Liang Yan and Bin Fan are co-first authors.) (Corresponding author: Hongmin Liu.)*

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Digital Object Identifier 10.1109/TGRS.2019.2958123

***Index Terms*— Domain adaptation (DA), pixel-level classification, self-training, triplet adversarial learning, very high resolution (VHR).**

I. Introduction

PIXEL-LEVEL classification for very high resolution (VHR) images is an enduring research task in remote sensing (RS) community that has great significance in infrastructure planning, land planning, and urban area change detection [1]–[3]. Existing methods [4]–[6] generally require lots of labeled samples to achieve promising performance. However, due to the diversity of RS image acquisition conditions [e.g., different spectral bands, color saturation, regions and ground sampling distances (GSDs)], it is time-consuming and laborious to provide annotated information for all images. Moreover, when applying the model trained with some labeled VHR images (source domain) to classify the VHR images acquired with different acquisition conditions (target domain), the performance may probably degrade. This phenomenon is known as domain shift [7]–[9] that is caused by the gaps between the distributions of different domains.

Distribution gapsbetween VHR data sets are mainly due to the diverse data acquisition sensors and varied data acquired regions. Specifically, in view of various data acquisition sensors, the obtained VHR images may be different in many aspects (e.g., spectral bands, color saturation, GSD, and shooting angles). Varied data acquired regions may result in significant distinctions in architectural style and urban layout between data sets (e.g., Fig. 4). Domain adaptation (DA) is a feasible way to solve the domain shift problem and its main purpose is to transfer the knowledge of the source domain to the target domain. This would benefit the performance of the learned classifier on the target domain. In this article, we focus on the study of the pixel-level DA method between VHR data sets, which is a significant but rarely studied problem.

To eliminate the domain shift, lots of methods are employed to pixel-level classification tasks on hyperspectral RS images. Banerjee *et al.* [10] proposed to construct a graph among categories and cope with DA problems by solving the graph-matching problem. Instead of matching graph properties across domains, Matasci *et al.* [11] proposed a semisupervised transfer component analysis method to map the distribution into a reproducing kernel Hilbert space and minimize the mapped distribution to eliminate the domain shift. Furthermore,

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Yang and Crawford [12] developed to eliminate the domain shift by aligning the distribution of the source and target data in the manifold space. By utilizing the feature extraction ability of deep learning [13], Zhou and Prasad [14] proposed a deep feature alignment neural networks method to employ deep convolutional recurrent neural networks to extract the discriminative features of the source and target domains and map the features layer by layer to a common space for alignment.

For VHR images, these methods may fail to extract semantic features and the context information for pixel-level classification tasks with a small number of spectral bands (e.g., RGB or IRRG). There are several studies that apply DA to VHR images, such as [15], [16]. Othman *et al.* [15] presented a deep convolutional neural network (DCNN) [17]-based DA method for VHR image classification. This method first utilizes DCNN to extract features, then exploits maximum mean discrepancy to minimize the distribution distance between the domains, and finally applies the Laplacian constrained graph matching method to preserve the geometry of the target domain data. Yan *et al.* [16] combined the global contrast-based salient region detection method with color-boosted and rotation invariant bag of visual words representation to transfer the parameters for cross-domain scene classification. However, these methods are proposed to address cross-domain tasks for scene classification, and may not be applicable to the densely prediction pixel-level classification task [18].

Recently, several DA methods have been proposed [19], [20] to deal with domain shift problem for pixel-level classification between street-view data sets based on the adversarial learning [21]. FCNs in the wild [19] addressed the domain shift problem in semantic segmentation by feature alignment across domains via adversarial learning. Sankaranarayanan *et al.* [20] proposed an approach based on generative adversarial networks [21] that projects the features to the image space using a generator. Then the discriminator operates on this projected image space. Those methods can be seen as the extension of domain adversarial neural networks (DANNs) [22], which is the first work that applies adversarial learning to solve domain shift problem on the image classification task, in semantic segmentation domain shift task.

Although previous adversarial-based DA methods [19], [20], [22], [23] can relieve the domain shift problem to some extent, there are several limitations. Firstly, they only consider the source and target domains, independently, in the process of adversarial training, where only the target domain information is used for learning a domain-invariant generator (i.e., the segmentation network in our task). In other words, only the target domain information is explicitly contributed to narrow the gap between the distributions of the source and target domains, as shown in Fig. 1(a). Secondly, they use the discriminator to classify whether a sample is from the source domain or the target domain. However, learning such a discriminator is harder than distinguishing whether two samples from the same domain or not. Such a strong but with some unnecessary information may be harmful to adversarial learning since the unnecessary information may mislead the generator in the learning process. Then, they use the intermediate features for adversarial learning to generate domain-invariant features and learn the classifier with only the source data. This is reasonable only when the domain shift problem is perfectly addressed in the feature level, otherwise, the learned classifier will no doubt suffer from the imperfect domain-invariant features. In addition, as pointed out in AdasegNet [24], feature-level adaptation for semantic segmentation domain shift tasks may suffer from the complexity of high-dimensional features, which contain many noise information. This increases the burden on the discriminator and makes the adversarial training process unstable [25]–[27]. Finally, they merely perform a global alignment across domains. The classwise discriminability for the target domain may not be guaranteed.

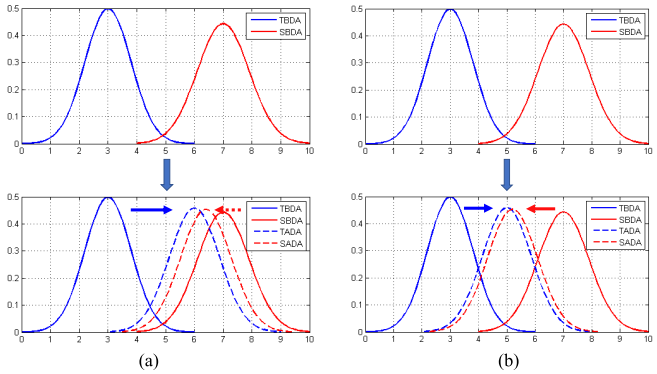


Fig. 1. Difference of underlying ideas between the traditional and the TriADA methods. (a) Traditional adversarial DA method only uses target data to optimize the feature extractor that the distribution of the source domain can only be implicitly brought closer to that of the target domain. (b) TriADA utilizes both domains' data to optimize the feature extractor (or classifier) that can explicitly push the distributions of the two domains being closer to each other. The solid arrows in blue and red indicate that the target data and source data are directly utilized to push their distributions being closer to each other. The red dotted arrow indicates that the distribution of the source domain is implicitly pushed closer to the target domain by using the target data.

To eliminate those limitations, we present a novel triplet adversarial DA (TriADA) method for addressing the domain shift problem in pixel-level classification of VHR images. The method consists of two modules: a semantic segmentation network **F** and a domain similarity discriminator **D**, as shown in Fig. 2. **D** and **F** are alternately optimized by playing a minimax game (also known as adversarial learning) as standard GANs [21] does. Different from previous methods, **D** takes a triplet of segmentation maps as input, where two segmentation maps from the same domain are to be distinguished from the two maps from the different domains in the process of adversarial learning. In this way, it simultaneously considers both the source and target domains' information, guiding the generator (i.e., **F**) to directly learn domain-invariant representations across domains during the adversarial learning. Compared to previous methods, it could be easier for our method to achieve the same distributions between the source and target domains as it explicitly uses information from both domains to narrow the distribution gap between domains, whereas previous works only explicitly consider the target domain, as shown in Fig. 1(b).

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Since the main effective information for DA is the information about the similarity of two domains, it is unnecessary to exactly classify which domain the input belongs to. Pursuing too hard for such ability (as previous works did) may introduce unnecessary information that may be harmful to the adversarial learning and produce a negative effect on DA. Our TriADA method is able to learn the similarity distribution of each pair of inputs by deciding whether two samples from the same domain or not. This specific implementation enables the generator to focus on generating domain-invariant features thus reducing the impact of unnecessary information. Moreover, we can utilize much more training samples by triplet adversarial learning, which can mitigate the overfitting in the case of insufficient training samples.

In addition, in contrast to [19], [20], we apply adaptation on the output space so that both the source and target data can be directly involved in the optimization of the classifier[1]. Moreover, since the output space shares more similarities in the spatial layout and local context [24] and distills the noise information, it is beneficial for the stable training of adversarial learning.

Finally, to improve the discriminability for each category in the target domain, we propose a simple but effective strategy called class-aware self-training (CAST). It is incorporated in the proposed adversarial DA method to generate pseudo-labels for target data based on the output of the discriminator. Meanwhile, to prevent the learned model biasing toward the initially well-transferred classes, we regard the predicted labels of target data with high adaptive confidence as pseudo-labels by category. These pseudolabels are utilized to retrain $\mathbf{F}$ so as to improve its discriminability for each category in the target domain.

In summary, the main contributions of this article can be summarized as follows.

1) A TriADA method is proposed for pixel-level classification of VHR images across different data sets. It simultaneously considers both the source and target domain information to learn a domain-invariant classifier and narrows the distribution gap between domains by explicitly using both domains' information in the process of adversarial learning.
2) A CAST strategy based on the output of the discriminator is proposed to treat the predicted labels of target data with high confidence as pseudolabels. These labels are utilized to retrain $\mathbf{F}$ to improve its discriminability in the target domain.
3) Extensive experiments have been conducted to show the effectiveness of the proposed method, including various ablation studies to support each design choice and motivation of our method, as well as the comprehensive comparisons to the existing competitors. These results on five DA situations generated by three VHR data sets clearly demonstrate the state-of-the-art performance of our method.

[1]Here the classifier means the last pixel-level classification layer in segmentation network.

The remainder of this article is organized as follows. Section II gives an overview of our method and introduces the notation used in this article. Section III presents the proposed method in detail. The data sets description and experiments are provided in Section IV. Finally, Section V draws the conclusion.

II. Overview of the Method

We denote the labeled source domain as D_S and the unlabeled target domain as D_T. In the problem of DA, we have access to source image $\mathbf{x_s} \in \mathbb{R}^{H\times W\times 3}$ with label $\mathbf{y_s} \in \mathbb{R}^{H\times W\times C}$ sampled from D_S, and unlabeled image $\mathbf{x_t} \in \mathbb{R}^{H\times W\times 3}$ sampled from D_T. H, W, C denote the height, width, and the number of categories, respectively.

The proposed TriADA eliminates domain shift in an end-to-end manner, which consists of two modules: a semantic segmentation network $\mathbf{F}$ and a domain similarity discriminator $\mathbf{D}$. Specifically, a triplet $\{\mathbf{x_{s,1}}, \mathbf{x_{s,2}}, \mathbf{x_t}\}$, which consists of two source images $\mathbf{x_{s,1}}$, $\mathbf{x_{s,2}}$ with labels and one unlabeled target image $\mathbf{x_t}$, is regarded as the input of $\mathbf{F}$. The generated three segmentation maps are upsampled to the size of $H \times W \times C$ followed by a softmax activation function to generate the corresponding probability maps, denoted as a triplet $\{\mathbf{P_{s,1}}, \mathbf{P_{s,2}}, \mathbf{P_t}\}$.

Similar to TripletGAN [29], we treat $\mathbf{P_{s,1}}$ as the anchor, and $\mathbf{P_{s,2}}$ and $\mathbf{P_t}$ are treated as the positive and the negative with respect to the anchor, respectively. The positive indicates that it is generated from the same domain as the anchor and the negative is the opposite. Among the triplet, the anchor and the positive pair $\{\mathbf{P_{s,1}}, \mathbf{P_{s,2}}\}$ are generated from the source domain and thus have labels. Therefore, they are used to compute a standard segmentation loss to optimize $\mathbf{F}$. Besides the two maps from the same domain, a triplet also contains a pair of maps from different domains, i.e., the anchor and the negative pair $\{\mathbf{P_{s,1}}, \mathbf{P_t}\}$. Our discriminator $\mathbf{D}$ is optimized to distinguish whether two maps in the input pair are from the same domain or not by learning with an adversarial loss defined on $\{\mathbf{P_{s,1}}, \mathbf{P_{s,2}}\}$ and $\{\mathbf{P_{s,1}}, \mathbf{P_t}\}$. Following [29], the pair $\{\mathbf{P_{s,2}}, \mathbf{P_t}\}$ contained in the triplet is not involved in the calculation of adversarial loss. Our experiments in Section IV-E1 also show that adding this pair is not necessary.

Since the adversarial training only performs an alignment across domains and $\mathbf{F}$ has not been trained with the labels of target data, the capability of $\mathbf{F}$ to discriminate each category of target data could be insufficient. Therefore, a CAST strategy is developed to assign pseudo-labels for target data, by which $\mathbf{F}$ is re-trained.

To sum up, the whole method is optimized by minimizing a multitask loss function as follows:

$$\min_{\theta_{\mathbf{F}},\theta_{\mathbf{D}}} L_{seg}(\mathbf{x_s}) + \lambda_{adv} L_{adv}(\mathbf{x_s}, \mathbf{x_t}) + \lambda_{st} L_{st}(\mathbf{x_t}) \tag{1}$$

where L_{seg} is the multiclass cross-entropy loss using labeled source images for semantic segmentation, L_{adv} and L_{st} denote the adversarial loss and the CAST loss, respectively. Here, λ_{adv} and λ_{st} are the weights to balance the corresponding loss, θ_F and θ_D denote the parameters of $\mathbf{F}$ and $\mathbf{D}$, respectively.

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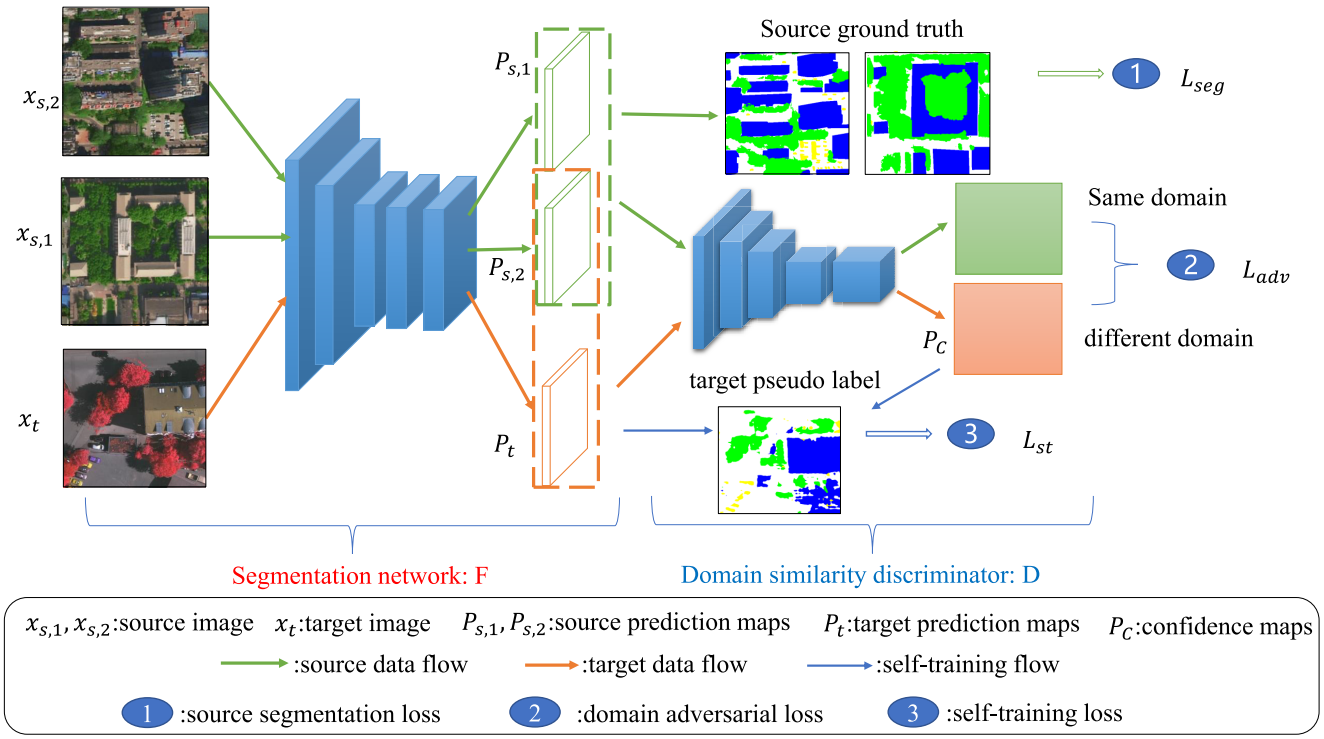


Fig. 2. Flowchart of the proposed method for DA. The architecture consists of two parts, basic network and domain similarity discriminator. The base network is a semantic segmentation network, such as DeepLab-v3 [28]. The domain similarity discriminator is composed of five convolutional layers and takes a pair of prediction maps as input to compute their similarity. Best viewed in color.

III. Domain Adaptation by Optimizing Over Multitask

As described in Section II, the proposed DA method is integrated into an adversarial learning framework by optimizing over different tasks simultaneously. These tasks include the standard semantic segmentation task for the source domain, a min–max game [21] between the source and target domains, and semantic segmentation task for the target domain through CAST. In the following sections, we elaborate on these tasks and their corresponding objectives, respectively.

A. Semantic Segmentation in the Source Domain

The purpose of semantic segmentation is to predict the category of each pixel in the input image. The existing DCNN-based [13], [30], [31] semantic segmentation methods [28], [32]–[38] have achieved great performance. We treat DeepLab-v3 [28] as the semantic segmentation network. In order to ensure that the network performs well on source images, which is known to be effective for the final semantic segmentation task, we optimize the standard supervised segmentation loss on the source domain. Therefore, a multiclass pixelwise softmax cross-entropy loss is utilized to achieve this goal

$$L_{seg}(\mathbf{x_s}, \mathbf{y_s}; \theta_\mathbf{F}) = -\mathop{\mathbb{E}}_{\mathbf{x_s}\sim D_S}\left[\sum_{i=1}^{N}\sum_{k=1}^{K}\mathbb{1}_{[k=\mathbf{y_{s,i}}]}\log f_F(\mathbf{x}_{s,i};\theta_F)\right] \tag{2}$$

where N is the number of pixels in each output probability maps, K is the number of categories. Here, $\mathbf{x_s}$ and $\mathbf{y_s}$ denote source image and the corresponding label, respectively. The indicator function $\mathbb{1}_{[k=y_{s,i}]}$ denotes that the value is 1 when $k = y_{s,i}$ and 0 otherwise. The function $f_F(\cdot)$ is the semantic segmentation network and θ_F is the corresponding parameters.

B. Triplet Adversarial Domain Adaptation

Before introducing our method, we simply review the previous adversarial based DA works [19], [22], [23]. To eliminate the domain shift, they play a minimax game between two networks. The discriminator is trained to distinguish which domain the input features are generated from. The feature extractor is trained to generate features to fool the discriminator. This process can be formulated as follows:

$$\begin{aligned} L_{adv,D}(f_F, f_D; \theta_D) = &-\mathop{\mathbb{E}}_{\mathbf{x_s}\sim D_S}[\log(f_D(f_F(\mathbf{x_s})))] \\ &-\mathop{\mathbb{E}}_{\mathbf{x_t}\sim D_T}[\log(1 - f_D(f_F(\mathbf{x_t})))] \end{aligned} \tag{3}$$

and

$$L_{adv,F}(f_F, f_D; \theta_F) = -\mathop{\mathbb{E}}_{\mathbf{x_t}\sim D_T}[\log(f_D(f_F(\mathbf{x_t})))] \tag{4}$$

where $f_F(\cdot)$ and $f_D(\cdot)$ denote the feature extractor and the discriminator, respectively. Here, θ_F and θ_D are the corresponding parameters. $L_{adv,D}$ is the discriminator loss trained to classify which domain the input features are generated from. $L_{adv,F}$ is an adversarial loss that optimizes the feature extractor to confuse the discriminator. According to [21], the algorithm can be converged by optimizing $\mathbf{F}$ and $\mathbf{D}$ alternatively and $\mathbf{F}$ is able to produce domain-invariant features.

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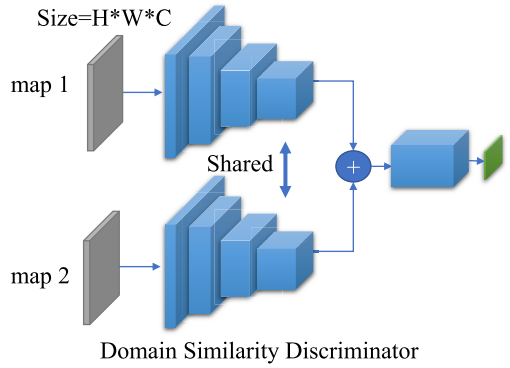


Fig. 3. Network structure of the proposed domain similarity discriminator. The first four convolutional layers are treated as domain feature extractor and the last convolutional layer is viewed as a domain similarity measure function. Color code used: gray = probability maps, blue = convolutional layers, and green = output result.

In previous adversarial-based DA works, only the target domain is explicitly used for learning a domain-invariant **F**. In other words, the distribution of the source domain can only be implicitly close to that of the target domain, by leveraging the optimization of the target domain for **F**, as shown in Fig. 1(a).

To simultaneously consider both the source and target domains into optimizing **F** to learn domain-invariant representations across domains during the adversarial learning, our TriADA develops a domain similarity discriminator (**D**). It takes a triplet of segmentation maps as input, where two maps from the same domain are to be distinguished from the two maps from the different domains in the adversarial learning, as shown in Fig. 2. In this way, our framework can explicitly take both domains' information into consideration to explicitly narrow the gap between the distributions of both domains, as shown in Fig. 1(b). Moreover, our triplet adversarial learning is easy to achieve the adaptation goal by deciding whether two samples from the same domain or not.

Specifically, to jointly consider both domains' information, the structure of **D** is shown in Fig. 3. The shared first four layers (Siamese networks) are viewed as a feature extractor to extract the domain information of the input maps. The outputs of the Siamese networks are concatenated on the feature dimension, and then input to another layer, which can be regarded as a domain similarity measurement function.

To promote **F** to generate domain-invariant results, **D** and **F** are integrated into an adversarial training framework. Firstly, we optimize **D** so that it can distinguish whether the input pair is from the same domain or not. Specifically, given two probability maps $\{\mathbf{P_{s,1}}, \mathbf{P_{s,2}}\}$ (or $\{\mathbf{P_{s,1}}, \mathbf{P_t}\}$) predicted by **F**, we forward them to **D** and optimize it by a binary cross-entropy loss $L_{adv,D}$. Then, 3 is replaced as follows:

$$\begin{aligned} &L_{adv,D}(f_D, f_F; \theta_{\mathbf{D}}) \\ &\quad = \underset{\mathbf{P_s}\sim f_F(\mathbf{x_s})}{-\mathbb{E}} [\log(f_D(\mathbf{P_{s,1}}, \mathbf{P_{s,2}}))] \\ &\qquad \underset{\mathbf{P_s}\sim f_F(\mathbf{x_s}),\mathbf{P_t}\sim f_F(\mathbf{x_t})}{-\mathbb{E}} [\log(1 - f_D(\mathbf{P_t}, \mathbf{P_{s,1}}))] \end{aligned} \tag{5}$$

where $f_D(\cdot)$ and $f_F(\cdot)$ represent the discriminator network and the segmentation network, respectively. θ_D denotes that we only update the parameters of **D**. We set the label of the pair as 1 when the two maps among it are generated from the same domain, and 0 otherwise.

Secondly, we reverse the label of the $\{\mathbf{P_{s,1}}, \mathbf{P_t}\}$ pair and minimize the adversarial loss $L_{adv,F}$. The adversarial loss in 4 can be written as

$$L_{adv,F}(f_D, f_F; \theta_{\mathbf{F}}) = \underset{\mathbf{P_s}\sim f_F(\mathbf{x_s}),\mathbf{P_t}\sim f_F(\mathbf{x_t})}{-\mathbb{E}} [\log(f_D(\mathbf{P_t}, \mathbf{P_{s,1}}))] \tag{6}$$

where θ_F denotes that we only update the parameters of **F**. The purpose of this loss is to spoof **D** by optimizing **F** to discriminate the $\{\mathbf{P_{s,1}}, \mathbf{P_t}\}$ pair as if they are generated from the same domain.

Through the domain adversarial training, **F** is able to produce domain-invariant predictions. However, since this process only aligns the distribution of different domains, the class-wise discriminative ability for the target domain may be insufficient. Therefore, we develop a CAST strategy to generate pseudo-labels for the target domain and retrain **F** with these pseudo-labels, which is described in Section III-C.

C. CAST for Target Domain

As mentioned above, although TriADA enables **F** to generate domain-invariant prediction results, it is less effective to discriminate different categories in the target domain since **F** does not "see" the ground truth of target data. Several methods [39]–[42] have been proposed to solve the classwise discriminative ability of the target domain. Long *et al.* [39] proposed a joint distribution adaptation (JDA) that jointly adapts both the marginal and conditional distributions in a principled dimensionality reduction procedure. In this way, JDA is able to increase the discriminability for the target domain. A tribranch method in [40] and [41] is developed to assign pseudo-labels to unlabeled target domain samples, which allow us to minimize the classification loss on the target domain. Zhang *et al.* [42] integrated both the supervised information from the source domain and the unsupervised low-rank property of the target domain into a robust classifier to assign pseudo-labels. According to those pseudolabels a transferable feature learning module is proposed to only match the distributions of the same class across domain so as to reduce the distributions discrepancy of the same classes and increase the distributions discrepancy of different classes across domains. Unfortunately, those methods are not suitable for assigning pseudolabels for the target domain under our adversarial learning framework.

To improve the discriminability for the target domain under our adversarial framework, a CAST strategy is developed to assign pseudolabels for the target domain, by which **F** is retrained. Specifically, during the adversarial training, the $\{\mathbf{P_t}, \mathbf{P_{s,1}}\}$ pair is fed into **D** and output a downsampled map. We upsample the output map to the same size of $H \times W \times 1$ followed by a sigmoid layer and denote it as $\mathbf{P_C}$. Each element in $\mathbf{P_C}$ represents the probability that the corresponding two pixels in the input pair belong to the

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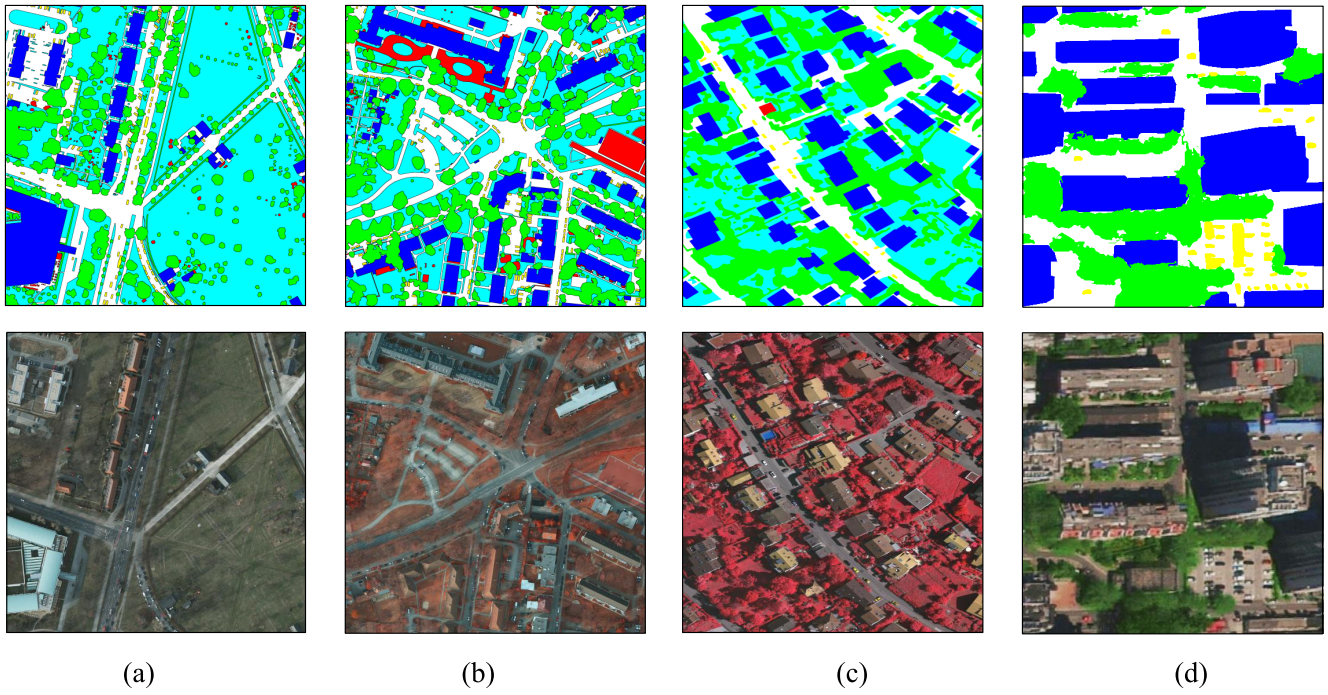


Fig. 4. Images and the corresponding ground truths sampled from three RS data sets. The first row denotes the ground truths, and the second row indicates original images. The Potsdam IRRG (POTIRRG), Potsdam RGB (POTRGB), and Vaihingen (VAI) include six categories: car (yellow), building (blue), tree (green), low vegetation (low veg, cyan), impervious surface (imp suf, white) and clutter/background (red). In addition, the black boundary, which is depicted for visual clarity. The clutter/background/boundary in POTIRRG or POTRGB data sets are ignored in the training process. Beijing data set has the same categories as the above data sets except that the categories of tree and low veg are both regarded as the category of tree.

same domain. Therefore, higher values in $\mathbf{P_C}$ indicate that the corresponding regions in $\{\mathbf{P_t}, \mathbf{P_{s,1}}\}$ are domain-invariant. In other words, the predicted results of these regions in $\mathbf{P_t}$ have higher reliability which could be selected as pseudolabels.

To avoid the learned model tending to initially well-transferred categories and ignoring other hard categories, we exploit $\mathbf{P_C}$ to select pixels with high confidence as pseudolabels category by category. We first obtain the label map $\mathbf{L_t}$ of the target image $\mathbf{x_t}$ by taking the index of the maximum value of each pixel in $\mathbf{P_t}$. Then, the confidence $\mathbf{S_{c,t}}$ of class c is picked from $\mathbf{P_C}$ according to the obtained labels in $\mathbf{L_t}$. Then, we sort $\mathbf{S_{c,t}}$ in a descending order and select the top $p\%$ elements denoted as $\mathbf{L_{c,t}}$. Note that some of the selected labels in $\mathbf{L_{c,t}}$ may have very small confidence in $\mathbf{P_C}$ which will damage the following retraining. To alleviate this problem, only the elements in $\mathbf{L_{c,t}}$ greater than a threshold T_{st} are selected as the label mask $\mathbf{M_{c,t}}$ of class c. Finally, we utilize $\mathbf{M_{c,t}}$ to choose the pseudo-labels $\hat{\mathbf{y}}_\mathbf{t}$ from the label map $\mathbf{L_t}$. The above procedure is summarized in Algorithm 1 and the CAST loss is formulated as follows:

$$L_{st}(\mathbf{x_t}; \theta_\mathbf{F}) = -\mathop{\mathbb{E}}_{\mathbf{x_t}\sim D_T}\left[\sum_{h,w}\sum_{c=1}^{C}\hat{\mathbf{y}}_t^{h,w,c}\log f_F(\mathbf{x_t}; \theta_\mathbf{F})\right] \quad (7)$$

where $\hat{\mathbf{y}}_\mathbf{t}$ denotes the pseudo-labels obtained by Algorithm 1. Through CAST strategy, we enable $\mathbf{F}$ to "see" some annotations of target data, ultimately improving the discriminative ability of $\mathbf{F}$ in the target domain.

Algorithm 1 Generation of Pseudolabels in CAST

Input: The confidence map $\mathbf{P_C}$ and the target predicted probability map $\mathbf{P_t}$, the proportion number p, and the threshold T_{st};

Output: The pseudo-labels $\hat{\mathbf{y}}_\mathbf{t}$ of target image $\mathbf{x_t}$;

1: Obtain the label map: $\mathbf{L_t} = argmax(\mathbf{P_t}, axis = 1)$;
2: Initialize the pesudo-labels to a map with the same size of $\mathbf{L_t}$: $\hat{\mathbf{y}}_\mathbf{t} = oneslike(L_t) \times 255$, where 255 denotes the pixel will be ignored when calculating the loss
3: **for** $c = 0; \quad c < C; \quad c++$ **do**
4: Pick confidence scores $\mathbf{S_{c,t}}$ of class c from $\mathbf{P_C}$ according to $\mathbf{L_t}$: $\mathbf{S_{c,t}} = \mathbf{P_C}\big[where(\mathbf{L_t} == c)\big]$;
5: Sort $\mathbf{S_{c,t}}$ in descending order and select the top $p\%$ elements as $\mathbf{L_{c,t}}$;
6: Select the label mask: $\mathbf{M_{c,t}} = where(\mathbf{L_{c,t}} > T_{st})$;
7: Generate the pseudo-labels of class c: $\hat{\mathbf{y}}_\mathbf{t}[\mathbf{M_{c,t}}] = \mathbf{L_t}[\mathbf{M_{c,t}}]$;
8: **end for**
9: **return** $\hat{\mathbf{y}}_\mathbf{t}$.

D. Training Step

To sum up, there are four learning objectives that should be optimized: 1) $\mathbf{F}$ should classify accurately on the source domain to ensure discriminative predicted results; 2) $\mathbf{F}$ is optimized to trick $\mathbf{D}$ to determine $\{\mathbf{P_t}, \mathbf{P_{s,1}}\}$ from different domains as if they are from the same domain so as to generate domain-invariant predictions; 3) $\mathbf{D}$ is optimized to distinguish whether the input pair is from the same domain to guarantee

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the discriminative ability of **D**; 4) After the three mentioned steps have been trained for T_N epochs, the segmentation ability of **F** for the target domain can be improved by minimizing L_{st}. To achieve these goals, the above processes are optimized as the following two steps.

Optimize F: In this step, the parameters of **F** are updated while the parameters of **D** are fixed. The network is optimized as follows:

$$\begin{cases} \min\limits_{\theta_F} L_{seg} + \lambda_{adv} L_{adv,F}, & if \quad n < T_N \\ \min\limits_{\theta_F} L_{seg} + \lambda_{adv} L_{adv,F} + \lambda_{st} L_{st}, & otherwise \end{cases} \tag{8}$$

where n represents the current epoch and T_N is the epoch to start employing CAST.

Optimize D: During this step, the parameters of **F** are fixed and **D** is updated. The learning objective is as follows:

$$\min_{\theta_D} L_{adv,D}. \tag{9}$$

During inference, since the discriminator network does not participate in the segmentation task, we remove it and only retain the segmentation network.

IV. Experiments and Analysis

Our experiments are conducted on three VHR data sets to show the effectiveness, superiority, and generalization ability of the proposed method. First, descriptions of the used data sets and experimental settings are presented. Second, qualitative and quantitative comparisons between existing methods and the proposed one are reported. Finally, the effectiveness of each part and the hyper-parameters utilized in our method are analyzed.

A. Data Sets Description

1) ISPRS Vaihingen Challenge Data Set: ISPRS Vaihingen Challenge Data Set (VAI) is a benchmark data set for the ISPRS Challenge in Vaihingen [43]. It consists of 32 3-band IRRG (Infrared, Red, and Green) VHR images taken by the airborne sensor from the city of Vaihingen, Germany, with a GSD of 9 cm. The size of each image is about $2\,500 \times 2\,000$. Only 16 of them are released with pixel-level labels, i.e., each pixel is classified into five valid categories: car, building, tree, low vegetation (low veg), and impervious surface (imp suf). We randomly select 10 images from the labeled images as training set, and the remaining six images as a testing set. In order to fit the model, we crop training set into a number of 512×512 patches with an overlap of 200 pixels (no overlap in testing set) in width and height, respectively. Finally, there are 2152 training images and 423 testing images in size of 512×512.

2) ISPRS Postdam Challenge Data Set: ISPRS Postdam data set (POT) [43], which has two different spectral bands, one is 3-band IRRG (POTIRRG), and the other is 3-band RGB (POTRGB). It consists of 38 VHR images with a size of $6\,000 \times 6\,000$ and the GSD is 5 cm, which is collected by airborne sensor from the city of Postdam, Germany. Only 24 images have pixel-wise labels, we randomly select 15 images as the training set, and the remaining 9 images as the testing set. It contains the same five useful categories as VAI data set. We crop the training set to a number of 512×512 patches with an overlap of 200 pixels (no overlap in the testing set). To this end, we produce 6000 images with a size of 512×512 for training and 1296 images for testing.

3) BeiJing City Data Set: The BeiJIng City data set (BEJ) is a set of satellite images with 30 cm GSD collected by ourselves on Baidu map. We collect a total number of 202 3-band RGB pixelwise labeled images with a size of about 1800×800. We crop all of them to a number of 512×512 patches without any overlap, and flip them horizontally and vertically. Finally, there are more than 4000 images, and we randomly sample 800 images as testing data and the rest as training data. Different from the above two data sets, we regard both tree and low vegetation as tree.

TABLE I

Setting of Hyperparameters λ_{adv}, p, T_{st}, λ_{st}, and T_N in Different Scenarios

parameters	λ_{adv}	$p(\%)$	T_{st}	λ_{st}	T_N
BEJ to POTRGB	0.01	20	0.4	0.01	3
BEJ to VAI	0.05	20	0.5	0.01	4
POTIRRG to VAI	0.01	20	0.6	0.01	6
POTIRRG to POTRGB	0.01	20	0.6	0.01	6
POTRGB to VAI	0.01	20	0.6	0.01	6

B. Experimental Settings

1) Network Structures: DeepLab-v3 [28] with ResNet-101 [31] pre-trained on the ImageNet [44] data set is regarded as the segmentation network. As illustrated in Fig. 3, the structure of the domain similarity discriminator is similar to [45], which consists of 5 convolution layers and the number of channels in each layer are {64, 128, 256, 512, 1}. The kernel size and stride of each layer is 4 and 2, respectively. Leaky-ReLU [46] activation function with a parameter of 0.2 is added after each convolution layer except the last layer and no batch normalization layer is applied.

2) Implementation Details: In this article, we use the PyTorch framework [47] to implement our method on a Titan XP GPU. Similar to [28], the segmentation network is trained by stochastic gradient descent (SGD) optimizer with Nesterov acceleration, whose momentum is set to 0.9 and weight decay is 5×10^{-4}. The initial learning rate is set as 2.5×10^{-4} and the polynomial decreased according to [34] and [35] with a power of 0.9. For the discriminator, we utilize the Adam optimizer with an initial learning rate of 1×10^{-4} and the identical decreasing strategy as training the segmentation network. The parameters of Adam are set to 0.9 and 0.99. Due to the memory limitation of GPU, we set the minibatch size of source domain as 2 and 1 for the target domain. The setting of hyper-parameters of λ_{adv}, p, T_{st}, λ_{st}, and T_N are reported in Table I, and more details are discussed in Section IV-F.

3) Evaluation Metrics: In order to quantitatively evaluate the semantic segmentation performance, two kinds of measurements are used, i.e., intersection over union (IoU) and

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F1 score (F1).

$$IoU(P_m, P_{gt}) = \frac{|P_m \cap P_{gt}|}{|P_m \cup P_{gt}|} \tag{10}$$

where P_{gt} is the set of ground-truth pixels and P_m is the set of predicted pixels. $\cap$ and $\cup$ denote $intersection$ and $union$ operations, respectively. $|\cdot|$ denotes calculating the number of pixels in the set

$$F1 = 2\frac{Pre \times Rec}{Pre + Rec}, \quad Pre = \frac{tp}{tp + fp}, \quad Rec = \frac{tp}{tp + fn} \tag{11}$$

where tp, fp, and fn are true-positives, false-positives, and false-negatives, respectively. Pre and Rec denote the precision and recall. Moreover, we treat clutter, background, and boundary as the same label in the ground truth, which are ignored when calculating the losses.

C. Comparing Methods

The proposed TriADA is compared with extensive state-of-the-art methods, including domain adversarial neural network (DANN) [22], adversarial discriminative DA (ADDA) [23], maximum classifier discrepancy (MCD) [48], adapt structured output space (AdasegNet) [24], and adversarial domain similarity network (ADDS) [49].

1) *Source-only* is a baseline model that only utilizes the source domain for training and directly tests the obtained model on the target domain, which should clearly exhibit the problem of domain shift.
2) *DANN* [22] is the first method that applies adversarial learning to solve domain shift problem on the image classification task. It performs a feature-level adaptation that takes the features generated before the classification layer as input to the discriminator. Moreover, DANN only considers the source and target domains, independently, when optimizing the adversarial learning scheme. To adapt DANN to semantic segmentation DA task, we use a segmentation network to replace the classification network.
3) *ADDA* [23] is similar to DANN and the main difference is that ADDA first pretrains source network with labeled source data and utilizes it to initialize the target network. During training, the parameters of source network are fixed, and only target network needs to be updated. We treat a segmentation network as the bone network of ADDA.
4) *MCD* [48] utilizes two task-specific classifiers as discriminators to align the distributions of each category between the source and target domains.
5) *AdasegNet* [24] considers that feature adaptation for semantic segmentation domain shift task may suffer from the complexity of high-dimensional features. They assumed that the output space contains similar spatial structure and local information for semantic segmentation task. Therefore, they applied the adaptation on the output space through adversarial learning.
6) *ADDS* [49] applies the adversarial domain similarity network to the feature space to jointly take dissimilar and similar information between domains into consideration.
7) *TriADA* is the proposed triplet adversarial DA method without applying the CAST module.
8) *TriADA-CAST* is our full method that employs the CAST module to TriADA.

To ensure the fairness of comparisons, all the evaluated methods apply the same segmentation network (Deeplab-v3 [28]) as the base network and are tuned to their best performance.

D. Domain Adaptation on VHR Remote Sensing Images

In this section, we extensively study different types of domain shifts on VHR data sets. Firstly, we attempt to transfer from BEJ to POTRGB with a large domain shift, which is jointly caused by various architectural styles across cities, large GSD difference between sensors and large saturation diversity. Secondly, we learn to transfer knowledge from BEJ to VAI with medium domain shift. The domain shift between them is caused by the difference between the regions where images are collected, and the difference between the spectral bands, saturation, and GSD. Third, we explore the adaptability between POTIRRG and VAI with a small domain shift, which can be seen as an adaptation between cities. Finally, we report the results of other two adaptation scenarios, i.e., POTIRRG to POTRGB and POTRGB to VAI, and discuss a failure case of the DA methods.

1) Large Domain Shift From BEJ to POTRGB: We begin to validate the proposed method by learning the large domain shift between BEJ and POTRGB data sets. As shown in Fig. 4, there are significant differences between them. During the training process, BEJ data set can be seen as the source domain and POTRGB as the target domain. Since tree and low vegetation are both considered as the category of tree in BEJ, we treat tree and low vegetation in POTRGB as one category (i.e., tree).

The results are presented in Table II, where the mean IoU and F1 score of the baseline method are 25.9% and 36.0%, respectively. After being processed by different DA methods, the segmentation performance of POTRGB is improved apparently, showing the necessity of DA. Our best model TriADA-CAST achieves mean IoU and F1 score as high as 46.3% and 57.3%, thereby improving the baseline by 20.4% and 21.3%, respectively. Compared with the other competing methods, the model TriADA still has higher performance. This demonstrates that the proposed TriADA is more beneficial to align the distributions. Moreover, the performance improvement after applying CAST proves the effectiveness of CAST module.

Specifically, the IoU for tree is only 8.5% in the baseline method, mainly due to the large difference in saturation between BEJ and POTRGB, as shown in Fig. 4. After adaptation, our method can achieve 59.3% IoU on tree, which is 5.0% higher than the second best one.

Fig. 5 shows the qualitative results where the source-only model suffers from a serious domain shift problem. The results of source-only usually appear as noisy segmentation or wrong context. After adaptation, this problem has been alleviated to a large extent. Compared with other methods, our method yields

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TABLE II

QUANTITATIVE COMPARISON (%) WITH THE STATE OF THE ART ON THE ADAPTATION SITUATION BEJ TO POTRGB, IMP SUF DENOTES IMPERVIOUS SURFACE, AND TREE DENOTES LOW VEGETATION AND TREE. THE VALUES IN BOLD ARE THE BEST

Method	car		building		tree		imp suf		Avg	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1	mean IoU	mean F1
Source-only	0.1	0.2	55.3	71.3	8.5	15.7	39.8	57.0	25.9	36.0
DANN [22]	**11.2**	**20.1**	53.3	69.5	32.5	49.1	43.2	60.3	35.0	49.8
ADDA [23]	0.4	0.8	62.1	76.6	54.2	70.3	51.2	67.8	42.0	53.9
MCD [48]	1.3	17.7	55.4	64.4	51.0	32.2	47.9	55.2	38.9	42.4
AdasegNet [24]	0.8	1.7	63.5	77.7	54.3	70.4	52.5	68.8	42.8	54.7
ADDS [49]	0.2	0.4	62.3	76.7	46.2	63.2	51.1	67.7	40.0	52.0
TriADA	0.3	0.5	64.6	77.8	54.4	72.5	57.0	73.3	44.1	56.0
TriADA-CAST	0.2	0.3	**65.3**	**79.0**	**59.3**	**74.5**	**60.3**	**75.3**	**46.3**	**57.3**

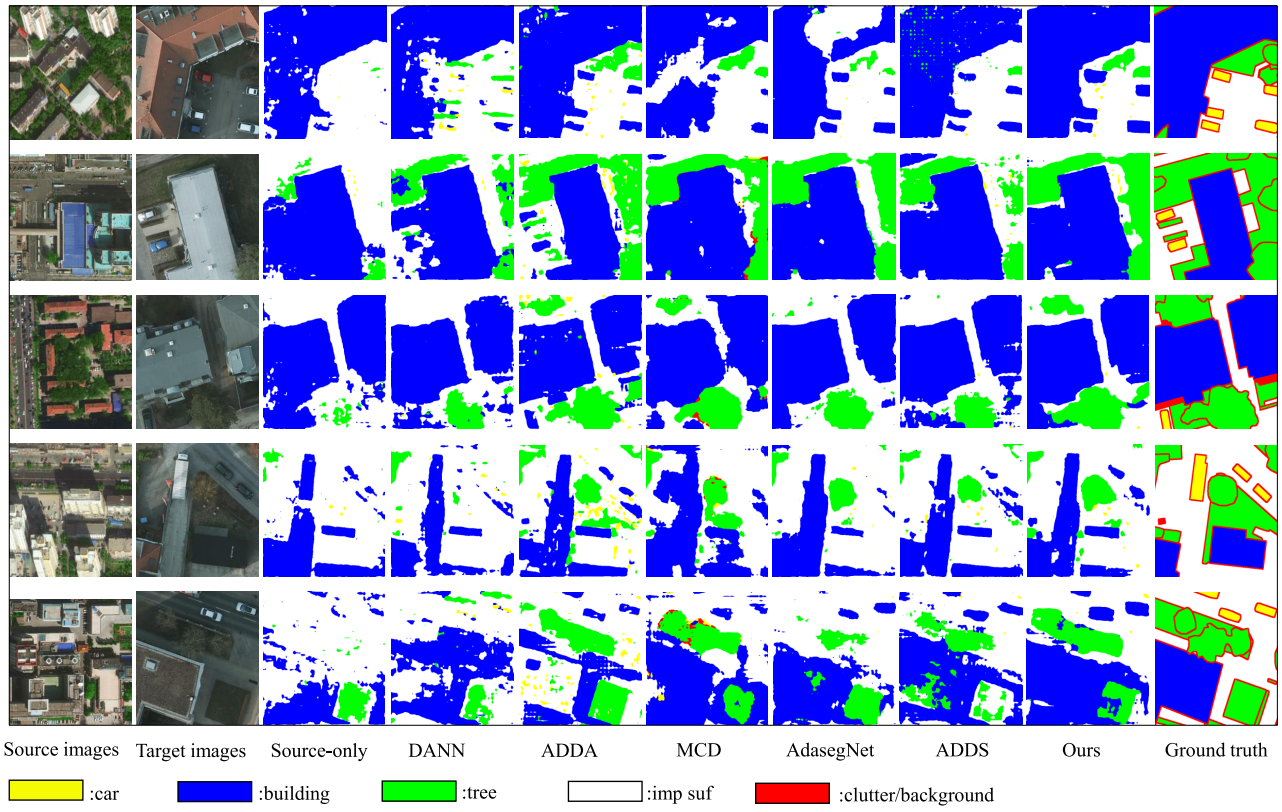


Fig. 5. Qualitative results of the adaptation situation BEJ to POTRGB. Each color represents, yellow: car, blue: building, green: tree, white: imp suf. Red denotes the boundary and the clutter/background, which are ignored in the training process.

better segmentation results, especially on the categories of imp suf and tree.

2) Medium Domain Shift From BEJ to VAI: Similar to the first experiment, we consider the categories of tree and low vegetation in VAI as the same label (tree) as well and treat BEJ as source domain and VAI as the target domain.

Table III describes the performance of our method and other competitive methods. The source-only method achieves very poor performance, with mean IoU and F1 score as 19.3% and 28.2%, respectively. For the category of tree, the IoU is even 0.0%. By observing the original data of the two data sets shown in Fig. 4, we find that the saturation between them is also very different and the color of tree is green in BEJ and red in VAI due to the different spectral bands used in these two data sets. Unsurprisingly, those differences directly lead to a sharp drop in performance.

The performance of TriADA is superior to other methods, and its mean IoU and mean F1 score achieve 48.6% and 61.0%. Moreover, the full method TriADA-CAST gives the best performance and significantly improves the mean IoU and F1 score of baseline from 19.3% and 28.2% to 52.6% and 64.6%, respectively. The IoU of tree category has increased from 0.0% to 68.1%, which is 41.3% higher than the second best one among all compared methods. Meanwhile, the IoU

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TABLE III

QUANTITATIVE COMPARISON (%) AGAINST THE STATE-OF-THE-ARTS ON THE ADAPTATION TASK BEJ TO VAI. THE VALUES IN BOLD ARE THE BEST

Method	car		building		tree		imp suf		avg	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1	mean IoU	mean F1
Source-only	2.3	4.6	30.8	47.1	0.0	0.2	44.0	61.1	19.3	28.2
DANN [22]	**19.3**	**32.3**	27.1	42.7	1.8	3.5	49.6	66.3	24.4	36.2
ADDA [23]	5.9	11.1	61.7	76.3	26.8	42.3	30.2	46.4	31.1	44.0
MCD [48]	9.7	17.7	47.5	64.4	19.2	32.2	38.2	55.2	28.6	42.4
AdasegNet [24]	10.3	18.7	66.5	79.9	9.6	17.5	34.3	51.1	30.2	41.8
ADDS [49]	18.3	31.0	61.9	76.5	12.0	21.4	38.3	55.4	32.6	46.1
TriADA	8.1	14.9	70.7	82.8	64.1	78.2	51.6	68.1	48.6	61.0
TriADA-CAST	13.0	22.0	**74.6**	**85.2**	**68.1**	**80.3**	**54.6**	**70.9**	**52.6**	**64.6**

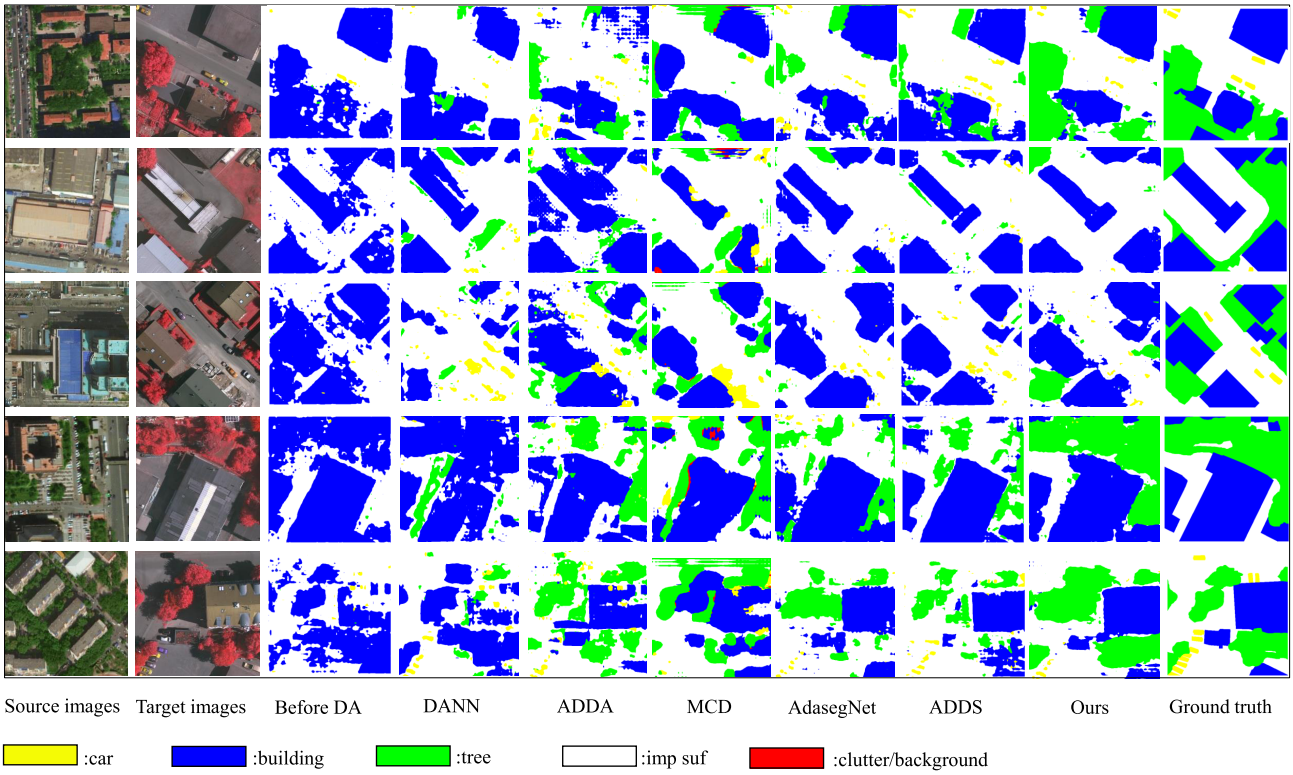


Fig. 6. Visual comparison of adaptation (BEJ → VAI) among our method and others. Each color represents, yellow: car, blue: building, green: tree, white: imp suf. Red denotes the boundary or the clutter/background, which are ignored in the training process.

of building category also increases from 30.8% to 74.6%. The above experimental results once more demonstrate the superiority and effectiveness of each part in our method.

The qualitative results are shown in Fig. 6. Similarly, the baseline model suffers from a severe domain shift problem. After adaptation, the segmentation results become much cleaner. Our method performs the best compared to other methods, especially for the categories of tree and building.

3) Small Domain Shift From POTIRRG to VAI: Since POTIRRG and VAI data sets have the same categories, we utilize all five valid categories in this experiment and treat POTIRRG as the source domain and VAI as the target domain.

As shown in Table IV, the mean IoUs and mean F1 scores for the proposed TriADA are much higher than those obtained by the baseline as well as other competitive methods. In addition, after employing CAST, the performance of our method is further improved. These results reaffirm the effectiveness of each proposed module. Specifically, the IoU of car is only 6% in the baseline, and our method can achieve a level as high as 46.7%, which is significantly better than other methods. Although the layout of imp suf and building between the two data sets vary greatly, our full method has increased their IoU from 27.5% and 44.6% of the baseline to 69.1% and 75.7%, respectively.

It can be seen from Fig. 7 that the segmentation network suffers from domain shift seriously on the target domain before DA. After DA, the segmentation performance for each category has been significantly improved. Compared with

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TABLE IV

QUANTITATIVE COMPARISON (%) WITH THE STATE-OF-THE-ARTS ON THE ADAPTATION BETWEEN POTIRRG AND VAI. THE VALUES IN BOLD ARE THE BEST

Method	car		building		tree		low veg		imp suf		Avg	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	mean IoU	mean F1
Source-only	6.0	11.4	46.6	63.5	42.0	59.2	23.9	38.6	27.5	43.1	29.2	43.2
DANN [22]	33.1	49.7	68.3	81.1	55.5	71.4	26.9	42.4	64.0	78.0	49.5	64.5
ADDA [23]	30.4	46.7	67.6	80.7	43.1	60.2	29.7	45.8	62.3	76.8	46.7	62.1
MCD [48]	8.3	15.3	52.9	69.2	32	48.4	25.3	40.5	55.0	71.0	34.7	48.9
AdasegNet [24]	35.6	52.6	68.7	81.5	54.8	70.8	33.7	50.4	65.0	78.7	51.6	66.8
ADDS [49]	38.0	55.1	70.8	82.9	53.3	69.5	29.6	45.7	65.6	79.2	51.5	66.5
TriADA	40.9	58.0	73.6	84.8	57.9	73.4	**35.0**	**51.8**	67.8	80.8	55.1	69.8
TriADA-CAST	**46.7**	**63.6**	**75.7**	**86.2**	**58.0**	**73.4**	34.5	51.3	**69.1**	**81.7**	**56.8**	**71.2**

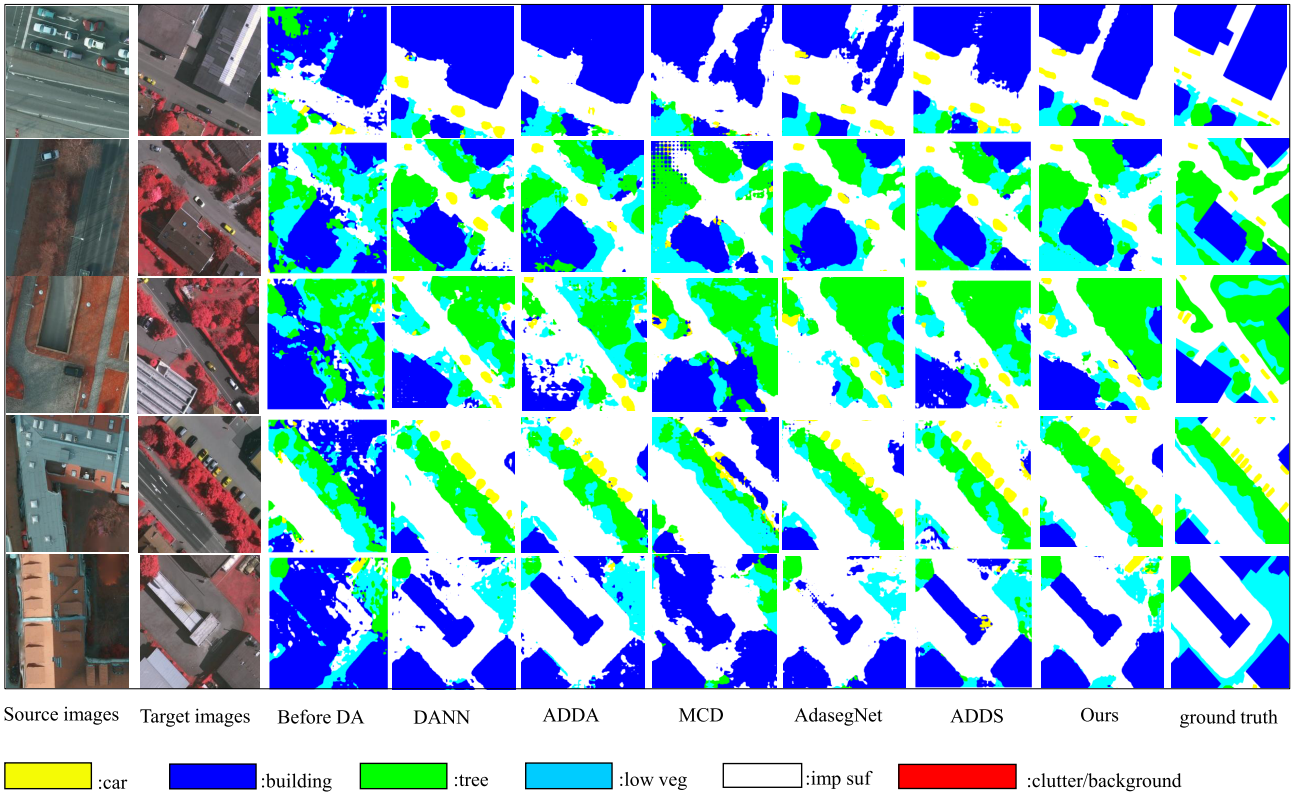


Fig. 7. Qualitative results show the adaptation between POTIRRG and VAI. Each color represents, yellow: car, blue: building, green: tree, cyan: low veg, white: imp suf. Red denotes the boundary or the clutter/background, which are ignored in the training process.

other methods, our method can preserve more details on edges, which demonstrates the superiority of the proposed method from a qualitative perspective.

4) Other DA Scenarios and Failure Case Discussion: To verify the generalization of our method, we construct other three adaptation situations, i.e., POTIRRG to POTRGB and POTRGB to VAI. The only difference between POTIRRG and POTRGB is the bands, and differences between POTRGB and VAI are reflected in the color bands, architectural style, and GSD. In addition, we also discuss some cases of poor adaptation performance.

Table V presents the comparison of our method and other state of the art. It shows that the adaptation situation of POTRGB to VAI suffers from a serious domain shift problem, whereas POTIRRG to POTRGB suffers less. An interesting phenomenon is that, although the domain shift suffered by POTIRRG to POTRGB is weak, its performance degrades after applying some DA methods, e.g., DANN [22] and MCD [48]. This phenomenon indicates that some DA methods will cause negative transfer if the domain shift is small, indicating a possible overfitting in learning domain-invariant features. In contrast, even though the performance of POTRGB to VAI is heavily influenced by the domain shift, all DA methods are improved compared to the baseline. This demonstrates the necessity of applying DA. The best performances achieved by our method in different scenarios demonstrate the effectiveness and generalization of our method.

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TABLE V

QUANTITATIVE COMPARISON (%) WITH THE STATE-OF-THE-ARTS ON THE ADAPTATION SITUATIONS OF POTIRRG TO POTRGB AND POTRGB TO VAI. THE VALUES IN BOLD ARE THE BEST

Method	POTIRRG to POTRGB		POTRGB to VAI	
Metric	mean IoU	mean F1	mean IoU	mean F1
Source-only	82.5	90.0	26.4	39.4
DANN [22]	79.5	88.3	35.4	50.3
ADDA [23]	83.9	91.0	40.8	53.1
MCD [48]	78.3	86.8	30.2	41.6
AdasegNet [24]	83.9	90.9	45.2	61.5
ADDS [49]	85.5	92.0	42.6	55.5
TriADA	86.9	92.8	49.7	65.6
TriADA-CAST	**88.0**	**93.4**	**51.4**	**66.5**

In DA, knowledge transfer is more difficult for the categories that are inherently difficult to segment, which could result in failure of domain adaption. In our experiments, the category of "car" could be of this case compared to other categories. This is because: 1) there are only several pixels occupied by a car making there is no enough context information for extracting good features; 2) the number of car pixels in the whole data set is much less than other categories. In other words, those reasons make the segmentation of car is difficult in RS images, especially in low-resolution images. It is even harder to deal with the domain shift for such category. These may be the reasons when adaptation from BEJ to VAI or POT usually cannot obtain satisfactory results for car (see Tables II and III). Since VAI and POT are high-resolution images where we can extract better features for car, adaptation of "car" between these two data sets does not suffer this problem heavily, at least not as serious as BEJ.

In addition, if the target and source domains have ambiguity in semantic classes, domain adaption is also hard to achieve a satisfactory result. This ambiguity will cause semantic label inconsistency across the source and target domains, and thus easy to result in failure of domain adaption. For example, in our experiments, we find that many cars are marked together in BEJ because of the low spatial resolution, whereas it is separated in POTRGB (it has enough GSD to distinguish cars from each other) as shown in Fig. 4.

E. Model Analysis

In this experiment, we present extensive studies on the situation of transferring from POTIRRG to VAI to investigate the evolution of the proposed method. To evaluate the effectiveness of each part of our method, we conduct extensive ablation experiments. We also analyze the capability of preventing over-fitting and the convergence of our model.

1) Evolution of the Method: To reveal the development of our method, abundant comparative experiments are conducted on the adaptation situation of POTIRRG to VAI and all the results are shown in Table VI. Specifically, L denotes the layer to perform concatenation in $\mathbf{D}$. The SC is the situation that the discriminator outputs a scalar by adding an averaging function in the last layer. This situation is the image-level adversarial. DP represents the discriminator utilized in this article that outputs a down-sampled map. PW is the situation that the discriminator outputs a pixel-wise probability map without upsampling. PW-CAST and DP-CAST present the results of applying CAST to PW and DP, respectively. The upsampling operation is only applied in DP-CAST. Note that since the output size of SC is a scalar, it is not suitable for applying CAST.

TABLE VI

EFFECTIVENESS OF VARIANTS ON DISCRIMINATOR IN THE SCENARIO OF POTIRRG TO VAI. THE VALUES IN BOLD ARE THE BEST

L	0	1	2	3	4
SC	53.0	52.4	52.9	52.3	52.5
PW	50.8	53.4	49.9	53.9	52.6
DP	55.7	54.8	52.9	55.3	55.1
PW-CAST	51.6	52.1	52.3	53.8	50.1
DP-CAST	56.5	56.5	55.7	56.2	**56.8**

By comparing SC and DP, it can be seen that employing a down-sampled map as the output of $\mathbf{D}$ is superior to output a scalar, no matter which layer to apply the concatenation. A possible reason is that the scalar output only aligns the distributions of the two data sets globally, whereas the spatial map output facilitates local alignment, which may be more suitable for pixel-wise classification. Compared with DP, the performance of PW is lower on all layer. The reason is that the receptive field of the pixelwise map in PW is smaller than that in DP, so that it does not take the context information into account, which is not beneficial to measure domain similarity. The large margin between PW-CAST and DP-CAST once again demonstrates that DP could benefit more than PW when applied with CAST. Compared with DP, DP-CAST not only achieves better performance, but is also more stable with different L, which indicates the effectiveness of the proposed CAST. According to those results, we utilize DP-CAST with L of 4 as the final discriminator, as shown in Fig. 3.

As discussed in Section II, when computing the adversarial loss, we can take $\{\mathbf{P_{s,2}}, \mathbf{P_t}\}$ into account in 5 and 6. In addition to the theoretical analysis in [29], we also conduct experiments to analyze whether it is necessary to add $\{\mathbf{P_{s,2}}, \mathbf{P_t}\}$ into the adversarial loss. The performance of adding $\{\mathbf{P_{s,2}}, \mathbf{P_t}\}$ in the model of TriADA is 54.5%, lower than that without adding it (55.1% acieved by TriADA). Similar performance degradation has been observed on TriADA-CAST (55.3% vs. 56.8%). Taking these reasons into consideration, the pair $\{\mathbf{P_{s,2}}, \mathbf{P_t}\}$ is not added in the adversarial loss.

2) Ablation Analysis: To evaluate the performance brought by each aspect, we focus on the situation of POTIRRG to VAI and conduct the ablation experiments. Table VII lists the performance of applying different parts. The baseline model means without utilizing any DA method. "FL" and "OL" denote the feature-level adaptation and the output-level

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TABLE VII

ABLATION ANALYSIS OF EACH PART IN OUR METHOD ON THE SITUATION OF POTIRRG TO VAI. "FL" DENOTES THE FEATURE-LEVEL ADAPTATION AND "OL" DENOTES THE OUTPUT-LEVEL ADAPTATION. "TRAD" AND "DSD" ARE THE TRADITIONAL DISCRIMINATOR AND THE PROPOSED DOMAIN SIMILARITY DISCRIMINATOR, RESPECTIVELY

Part	FL	OL	TraD	DSD	CAST	mean IOU
baseline						29.2
Method	✓		✓			49.5
	✓			✓		51.5
		✓	✓			51.6
		✓		✓		55.1
		✓		✓	✓	**56.8**

TABLE VIII

EXPERIMENTAL RESULTS (MIOU %) OF POTIRRG TO VAI USING DIFFERENT AMOUNTS OF TARGET IMAGES DURING TRAINING. NOTE THAT WE COMPARE TRIADA TO ADASEGNET, AND ALL SOURCE DATA FROM POTIRRG ARE USED DURING TRAINING

Target percent	1%	5%	10%	25%	50%	100%
AdasegNet [24]	36.7	45.9	44.5	50.2	49.4	51.6
TriADA	46.6	53.5	53.3	53.5	53.6	55.1

adaptation, respectively. "TraD" and "DSD" are the traditional discriminator used in [24] and the proposed domain similarity discriminator, respectively.

By comparing "FL+TraD" and "OL+TraD," it is clear that applying adaptation to the output-level achieves better result than that to the feature-level. The superior performance of "OL+DSD" over "FL+DSD" once again indicates the superiority of applying adaptation to the output space, no matter which kind of discriminator is used.

Compared to "FL+TraD" ("OL+TraD"), "FL+DSD" ("OL+DSD") achieves nonignorable improvement, especially when applying adaptation to the output level. This indicates the superiority of our proposed DSD over traditional adversarial discriminator. Finally, the best performance achieved by 'OL+DSD+CAST' verifies the effectiveness of the proposed CAST module.

3) Analysis of Different Amounts of Target Data: To further explore the capability of preventing over-fitting of our model, we conduct an experiment with extremely insufficient target data under the adaptation situation of POTIRRG to VAI. During training process, all images in the POTIRRG data set are used, while different numbers of target images in VAI are randomly sampled. As shown in Table VIII, our method can achieve comparable performance even with only 5% of the target data, and only 1.6% worse than the model using full target images. In contrast, the performance of AdasegNet drops dramatically as the target data decreases. These indicate that our method is applicable with extremely insufficient data and has the ability to mitigate overfitting.

4) Convergence Analysis: To verify the convergence of the adversarial training, we plot the curves of **F** loss and **D** loss, as shown in Fig. 8(b). For comparison, the curves of

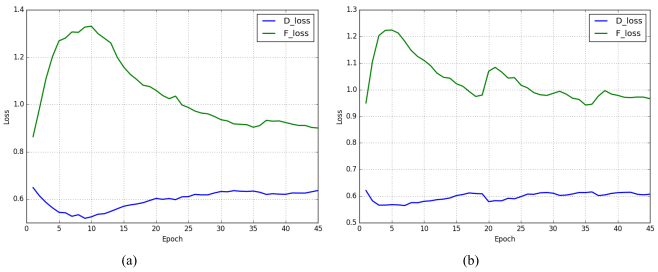


Fig. 8. Curves of adversarial training loss (D loss and F loss). (a) AdasegNet [24] is the compared method that is similar to ours. (b) TriADA is the proposed triplet adversarial DA method.

TABLE IX

SENSITIVITY ANALYSIS OF λ_{adv}. THE PERFORMANCE DENOTES THE MEAN IOU (%)

λ_{adv}	0.0001	0.001	0.01	0.05	0.1
BEJ to POTRGB	42.9	43.2	**46.3**	44.6	41.5
BEJ to VAI	48.3	38.9	50.9	**52.6**	50.4
POTIRRG to VAI	53.4	54.8	**56.8**	55.6	51.8

AdasegNet [24] (one competitive method to ours under the same framework of adversarial based DA) are also shown here, Fig. 8(a). Both of the two figures demonstrate a clear trend of convergence. Note that for adversarial learning, the target is not to minimize/maximize the loss, but for making two losses being stable, i.e., achieving a balance state.

F. Hyperparameters Sensitivity Analysis

In addition, to study the influence of parameter settings on our method, we present the sensitivity analysis of the following hyper-parameters: 1) the balance weight of adversarial loss λ_{adv}; 2) the proportion number p; 3) the threshold T_{st} to select pseudo-labels; 4) the balance weight of CAST loss λ_{st}; and 5) the epochs T_N to start CAST.

1) Influence of the Weight of Adversarial Loss: Table IX exhibits the mean IoU obtained by selecting the hyper-parameter λ_{adv} in $\{0.0001, 0.001, 0.01, 0.05, 0.1\}$. These results show that setting λ_{adv} between 0.01 and 0.05 leads to better performance. Giving more power to the adversarial loss, i.e., set λ_{adv} as 0.1, is harmful to the performance of our model. The performance degradation when λ_{adv} is too small indicates the importance of the adversarial loss.

2) Parameters Analysis of CAST: As described in Section III, there are four parameters in CAST, i.e., p, T_{st}, λ_{st}, and T_N, needed to be set. The results in Fig. 9(a) illustrate that better performance is achieved when p is set as 20% on all adaptation scenarios. The performance is relatively poor when p is either too small or too large. On the one hand, for a very small p, the pseudolabels generated in CAST are not enough to train target data. On the other hand, the pseudolabels generated by a very large p inevitably contain many incorrect labels that lead to inferior performance.

To investigate the impact of T_{st} on CAST, we conduct a series of experiences with T_{st} in the range of [0 0.7]. As depicted in Fig. 9(b), the peak performances are obtained

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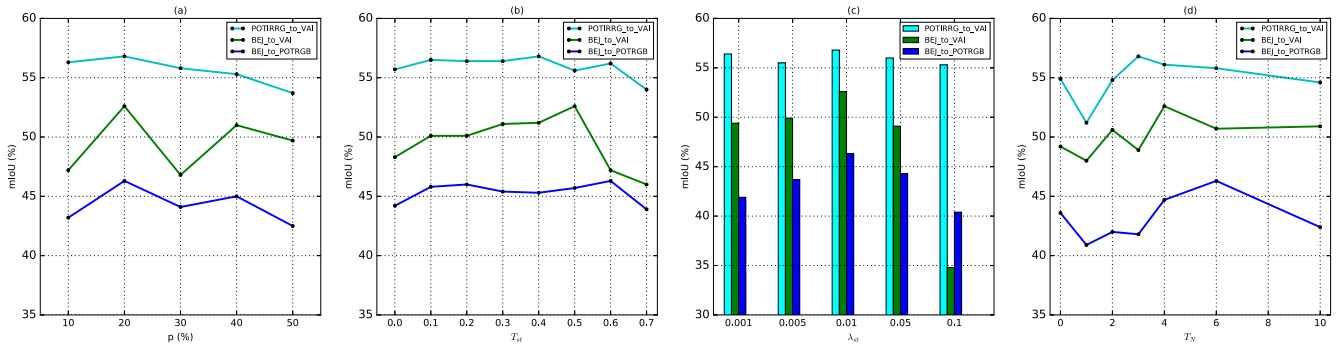


Fig. 9. Sensitivity analysis for the parameters used in CAST. (a) Proportion number p. (b) Pseudolabels Threshold T_{st}. (c) Balance weight of CAST loss λ_{st}. (d) Epochs of starting CAST T_N.

when T_{st} is set as 0.4, 0.5, and 0.6 for different adaptation scenarios (POTIRRG to VAI, BEJ to VAI and BEJ to POTRGB), respectively. A conceivable reason for the three scenarios to take different T_{st} is that for the situation of larger domain shift, the predicted results of $\mathbf{F}$ is not confident enough in the aspect of domain-invariant, and thus a larger T_{st} is needed to select pseudo-labels with high confidence. However, T_{st} cannot be too large (e.g., larger than 0.7) as it leads to less pseudolabels for retraining. On the other hand, too small (e.g., 0.0) T_{st} cannot filter out pixels with poor domain-invariant property.

We choose λ_{st} in $\{0.001, 0.005, 0.01, 0.05, 0.1\}$ to reveal the importance of CAST. Fig. 9(c) shows that our model performs better when λ_{st} is between 0.005 and 0.05, and the best performance is achieved when λ_{st} is set as 0.01. The larger power of CAST (e.g., $\lambda_{st} = 0.1$) results in decreased performance. This is because there are inevitably erroneous labels in the generated pseudo-labels.

Since the domain-invariance of the predicted results cannot be guaranteed during the initial training stages, we set up experiments to verify the appropriate epoch to start applying CAST. As shown in Fig. 9(d), the scenarios with larger domain shift achieve the best performance after training more epochs. It is mainly because the domain shift between data sets with huge differences is difficult to eliminate. Intuitively, when CAST is applied too early (e.g., T_N is set as 1 or 2) the performance is poor, indicating that the domain-invariance of the model is weak at the early stage. Furthermore, the results signify that applying CAST too late (e.g., T_N is set as 10) is also not conducive to improve the performance.

V. Conclusion

We have proposed a novel end-to-end TriADA method to solve the domain shift problem in pixel-level classification between VHR data sets. Different from previous adversarial DA methods that independently consider the source and target domain during the adversarial learning such that only the target domain is explicitly used for narrowing the domain gap, our method jointly considers both domains to learn a domain-invariant pixel-level classifier in the process of adversarial learning. Benefited from the proposed domain similarity discriminator, our method can explicitly leverage both the source and target domains to narrow the distribution gap between domains by adversarial learning. To further improve the discriminative ability of the classifier for each category in the target domain, a CAST strategy is developed, which is incorporated in the proposed adversarial DA method to generate pseudolabels based on the output of the discriminator. Extensive experiments have validated the effectiveness and generalization of the proposed method. In the future, we plan to further improve this method from the following aspects: 1) proposing a suitable method for adaptation at multiple feature layers; 2) introducing reasonable unsupervised constraints for target data; and 3) proposing a more effective self-training strategy.

Acknowledgment

The authors would like to thank the Reviewers and Editor who provided many valuable comments and suggestions, which improve this article greatly.

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